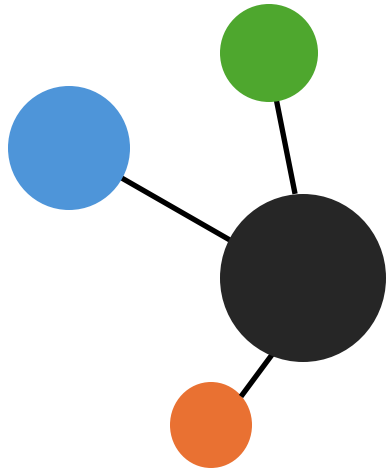


Clustering for Cardiovascular Risk Prediction in Type 2 Diabetes



Machine Learning Derived Phenotypes for Precision Management and Personalized Risk Stratification

Research Team: Alisha Pham, Yunjing Wei, Erin Liu

Department Advisors: Marina Tawfik, Jennifer Campbell

THE CLINICAL IMPERATIVE



The Global Epidemic

Type 2 Diabetes (T2D) affects over **747 million** people worldwide

T2D patients face **2-4x higher** cardiovascular risk compared to non diabetics

The Research Gap

Existing model that integrates routine clinical data achieves a **moderate C-index** of 0.74

No study has combined both routine clinical data and **clustering** to stratify cardiovascular risk in T2D

RESEARCH QUESTIONS



1. Can unsupervised clustering of EHR and 12-lead ECG data identify clinically meaningful phenotypic subgroups in Type 2 Diabetes patients?

2. Do these subgroups improve Major Adverse Cardiovascular Events predictions at 1, 3, and 5 years?

METRICS



F2 Score

Main Metric

$$F_2 = \frac{5 \cdot \text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}}$$

Clinical Priority: Recall over Precision

The F2 Score weighs Recall twice as heavy as Precision. In clinical deployment, the cost of missing a high risk patient carries a higher penalty than a false alarm.

Recall

The ability to correctly identify patients with high risk

Precision

Quantifies the accuracy of high risk flags

Specificity

Measures how accurately the model flags truly low risk patients

C Index

Measures how well the model ranks higher risk patients above lower risk patients

METHODOLOGY OVERVIEW



1



DATA PIPELINE

2



LATENT PHENOTYPING

3



CLINICAL VALIDATION

PREPROCESSING & FEATURE SELECTION



1 Partitioning & Sparsity

60/20/20 Split
Stratified by MACE. Re-sampling enforced if feature missingness difference >5% across sets

Sparsity Filtering
Features with >40% missingness in train were excluded

2 Imputation & Scaling

Unsupervised MICE

Feature Set. Bayesian Ridge

Skewness Correction

Applied optimal transformation (log1p, sqrt or identity) based on maximal skewness magnitude reduction to skewed features

Z Score Scaling

Continuous features standardized

3 Dual Gate Feature Selection

Continuous Pipeline

Excluded low variance features (SD < 0.01). Tested Mann Whitney U test (Symmetric) or Welch's t-test (Nonsymmetric)

Binary Pipeline

Dropped imbalanced features (Prevalence <5% or >95%). Chi squared test.

Significance Threshold

Retained features where $p < 0.05$

Data Leakage Prevention

All transformations, imputation and feature selection were established on training and then applied to validation and test.

THE OPTIMIZATION FRAMEWORK



Hyperparameter Grid

NC: 5,10,15
NN: 30, 50, 75, 100
MD: 0
K: 3, 4, 5

All hyperparameter combinations evaluated iteratively

1. Dimension Reduction
For every hyperprameter combination fit **UMAP** on feature training and transform validation.

2. Clustering
Apply **KMeans** to train UMAP embeddings. Exclude configuration if any training cluster size <2%. Assign validation to nearest cluster.

3. Prognostic Modelling
Fit **3 Logistic Regression** models (1, 3, 5 years) with selected features and cluster membership on train across a **0.05 threshold sweep**.

4. Performance Analysis
For each hyperparameter combination and logistic regression model record the highest F2 Score on validation and their corresponding threshold.

Optimal Configuration

Max F2 Score

Maximizes average validation F2 Score across 1, 3, 5 year time horizons

COMPARATIVE EVALUATION FRAMEWORK



Baseline Architectures

Selected Features

VS

Augmented Architectures

Selected Features

+ Cluster Membership From Optimal Configuration

Cluster Membership Variables: (k-1) Binary Variables The Cluster with the Lowest MACE Rate corresponds to the Baseline

1. Risk Classification
Logistic Regression

Optimal threshold for both architecture tuned on validation

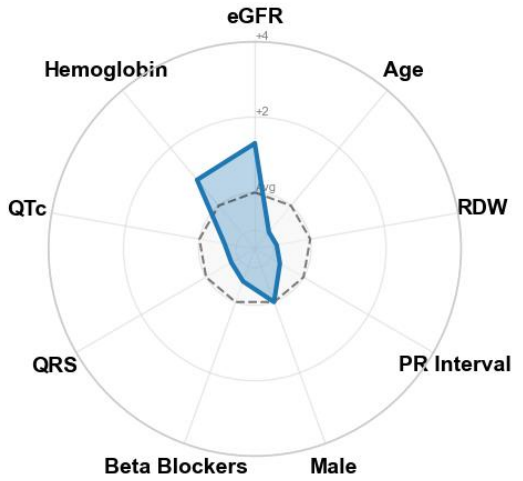
Reports Precision, Recall, Specificity and Accuracy across 1, 3 and 5 years

2. Survival Validation
Cox Proportional Hazards Model

Reports C Index

PHENOTYPE DESCRIPTION

Cluster 0: Low Risk Phenotype



Healthy Kidney Function

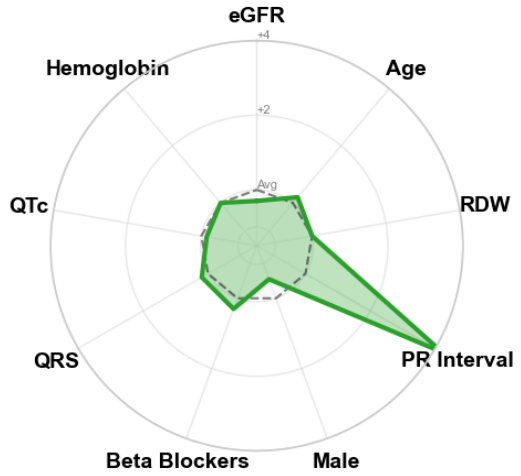
Favorable Hematologic Profile

Normal Cardiac Conduction

Younger Age

Lower Cardiovascular Medication Burden

Cluster 1: AV Conduction Delay Phenotype



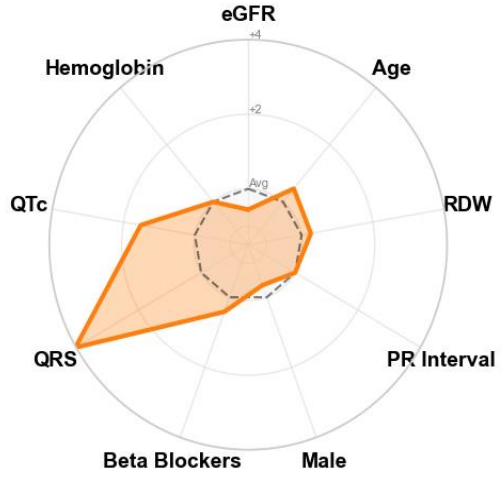
High PR Interval Prolongation

High First Degree AV Block Prevalence

Increased Use of AV Nodal Slowing Medications

Female Dominated

Cluster 2: Ventricular Electrical Remodeling Phenotype

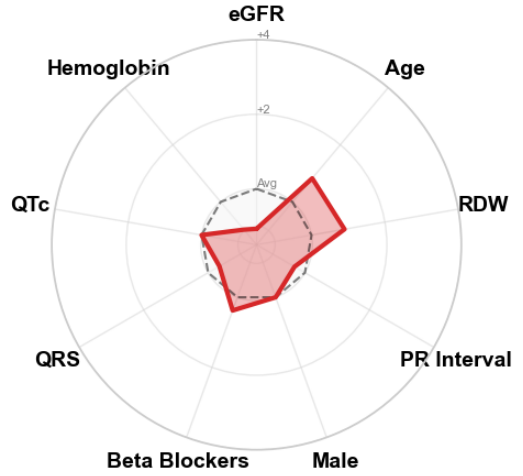


Severe Prolonged QRS: Ventricle Polarization Delay

Prolonged QTc: Ventricle Depolarization and Repolarization Delay

Increased RBBB / LBBB

Cluster 3: Cardio Renal Anemia Phenotype



Reduced Kidney Function

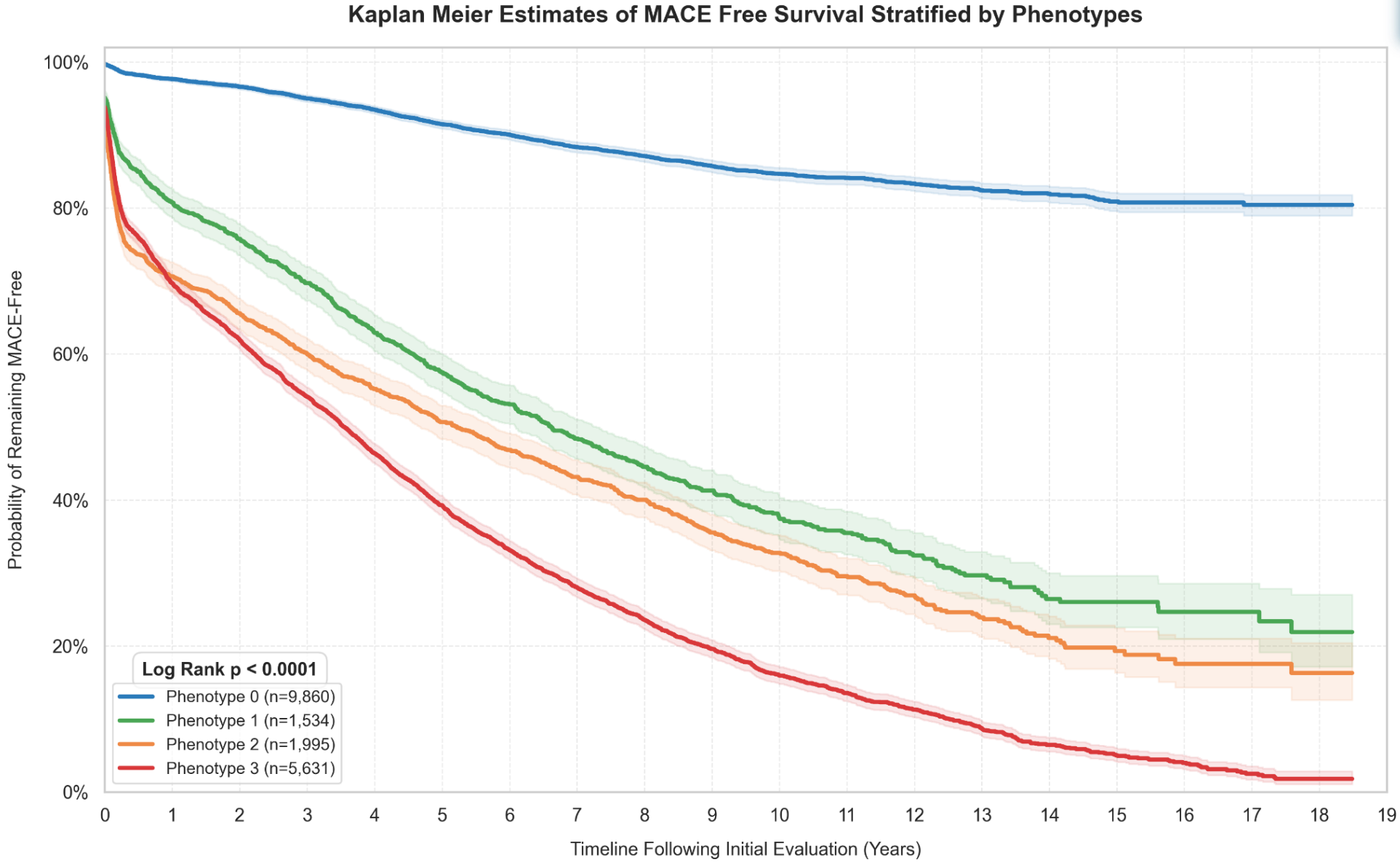
Anemia

Older Age

Cardiovascular Disease Burden: Heartfailure, Atrial Filbriation

Increased Cardiovascular Treatments

KAPLAN MEIER CURVE



METRIC RESULTS



Risk Classification Results

Metric	Year 1		Year 3		Year 5	
	Base	Aug	Base	Aug	Base	Aug
Precision	0.3757	0.3853	0.4939	0.4988	0.5933	0.6037
Recall	0.8770	0.8638	0.9057	0.9047	0.9467	0.9391
Specificity	0.7463	0.7601	0.7026	0.7086	0.6333	0.6516
Accuracy	0.7657	0.7754	0.7518	0.7562	0.7464	0.7554

Survival Validation Results

Metric	Base	Augmented
C Index	0.8429	0.8443

All hyperparameters were locked after Validation. Final performance is reported exclusively on the unseen Test set.

CONCLUSIONS



Clinically Distinct Phenotypes
Unsupervised clustering identified phenotypes in Type 2 Diabetes patients with significant different MACE risk

1. Low Risk Phenotype
2. AV Conduction Delay Phenotype
3. Ventricular Electrical Remodeling Phenotype
4. Malignant Cardio Renal Anemia Phenotype

Cluster Membership Did Not Improve Risk Prediction
Base and augmented models demonstrated strong and comparable predictive performance across all time horizons

Strong Predictive Performance
Both models achieve a C Index of 0.84, outperforming the 0.74 C Index reported in prior studies

LIMITATIONS



DATA LIMITATIONS

Single Centre Dataset
Phenotypes may not generalize to other populations

Static Phenotyping
Model uses index visit data only, Does not capture phenotype switching

Loss to Follow Up
Patients failed to follow up may have had MACE shortly after their last visit

METHOD LIMITATIONS

Missingness Bias
Features with >40% missing data were removed, Highly correlated with substantial missingness features were excluded

Metric Selection Bias
Hyperparameters and thresholds were optimized using the Fβ metric with β = 2. Alternate β values may provide a better balance between false positives and false negatives.

Limited Regularization Tuning
Regularization parameters were not optimized, which may reduce model generalizability and increase risk of overfitting or underfitting.

FUTURE DIRECTIONS



Nonlinear Risk Prediction
Neural networks could capture complex nonlinear relationships and feature interactions, enabling more accurate risk prediction

Time Dependent Data Modelling
Sequential models such as recurrent neural networks could incorporate longitudinal patient data to better model changes in risk over time

Sparse Predictive Features
Include highly correlated and missingness features (NT-proBNP peri highest, direct bilirubin peri)

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